GENDERTAGGER: Bridging the Gender Data Gap in Digital Editions through Multi-Method Annotation Workflows

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Abstract: Digital scholarly editions frequently rely on person registers to reconstruct historical networks and contextualize sources. However, gender information in such registers is often incomplete, making structural imbalances within historical datasets difficult to analyze. To address this challenge, we developed genderTagger, a web-based application for semi-automated gender annotation of TEI/XML person registers.

The system combines multiple classification approaches, including authority data queries, heuristic methods, statistical models, Transformer-based classifiers, and locally operated language models. Rather than relying on fully automated classification, genderTagger integrates these methods into a human-in-the-loop workflow that supports editorial review, uncertainty-aware decision-making, and direct reintegration into existing TEI/XML workflows.

In addition to a tabular annotation interface, the tool provides a gamified mode with score tracking and leaderboards to support collaborative metadata curation. Using the schleiermacher digital edition as a case study, the paper discusses methodological challenges of automated gender annotation in historical corpora, including incomplete authority data, ambiguous naming practices, and algorithmic bias.

The project demonstrates how interoperable and uncertainty-aware research software can support Digital Humanities workflows while critically engaging with the limitations of computational gender classification.

Keywords: gender data gap, digital humanities, digital scholarly editions

1 Introduction

Digital scholarly editions rely heavily on structured metadata and person registers to contextualize historical actors and reconstruct intellectual, social, and institutional networks. In many editorial projects, however, metadata concerning gender remains incomplete or entirely absent, making structural imbalances within historical corpora difficult to analyze. This lack of representation contributes to the broader phenomenon commonly described as the gender data gap. **[FEHLT: Literatur zum Gender Data Gap und feminist DH ergänzen]**

At the same time, the increasing scale of digital editions makes manual metadata enrichment increasingly time-consuming. Automated approaches to gender classification based on personal

names therefore offer promising possibilities for supporting editorial workflows. However, such approaches remain methodologically challenging in historical contexts due to ambiguous names, inconsistent spelling variants, incomplete authority data, and the risk of reproducing historical and algorithmic biases. **[FEHLT: konkrete Beispiele oder Literatur zu Problemen historischer Namensklassifikation]**

To address these challenges, we developed *genderTagger*, a web-based application for semi-automated gender annotation of TEI/XML person registers. The system combines multiple classification approaches—including authority data queries, heuristic methods, statistical models, Transformer-based models, and local large language models—within a human-in-the-loop workflow that preserves editorial oversight and uncertainty-aware review processes. **[FEHLT: klarer Research Gap – warum bestehende Tools/Ansätze nicht ausreichen]**

In addition to supporting systematic metadata enrichment, genderTagger introduces a gamified annotation mode designed to increase motivation and annotation speed during collaborative curation tasks. **[FEHLT: Motivation für Gamification + evtl. Literatur zu Annotation/Crowdsourcing/Gamification]**

Using the *schleiermacher digital* edition as a case study, this paper presents the system architecture, methodological design decisions, and initial observations regarding workflow integration and classification agreement. **[FEHLT: explizite Contributions als Bulletpoints oder klarer Absatz]**

2 Study Design

We treat the enrichment of a TEI/XML person register with gender annotations as a comparative methods study rather than as a software-engineering exercise. Three constraints typical of digital scholarly editions shape the design: (i) registers contain heterogeneous entries, with some carrying GND identifiers but most relying only on a name string and biographical dates; (ii) no single name-based classifier covers the spectrum of historical, multilingual, and ambiguous forms encountered in such a register; and (iii) editors need agreement and disagreement between methods exposed as a triage signal, not collapsed into a single opaque label.

The study is carried out on the Schleiermacher Digital person register (§[TODO: dataset section]) and is implemented as a reproducible Python pipeline of independent scripts, each consuming a CSV of the register and producing a CSV of per-method results, together with a small interactive review prototype used to validate selected entries against the machine output.

2.1 Components

The pipeline has three stages, each producing tabular artefacts that the next stage consumes.

Data preparation. The source TEI/XML register is flattened into a single table with one row per person and the editorially relevant fields per entry — identifier, surname and forename(s), vital dates, free-text biographical note, alternative name forms, and a GND identifier where present. Entries that carry a GND identifier are additionally resolved against the authority record, yielding a partial ground-truth table used both as a classifier in its own right and as the evaluation reference for the others.

Classifier bank. Each gender-classification method is applied independently to the prepared register and produces, per entry, a label drawn from a small shared vocabulary (`male`, `female`, `unknown`, with `institution` where the method can express it) together with a method-specific confidence score. The methods span five families — authority lookup, dictionary heuristics, statistical name models, transformer classifiers, and a locally hosted LLM — and are described individually in §3. Keeping the methods independent rather than chaining them keeps the comparison clean: any single method can be re-run, replaced, or removed without disturbing the others, and the slow methods can be run once and re-aggregated freely.

Aggregation, agreement, and review. Per-method labels are joined into a single table from which a combined label, the number of methods that returned a substantive answer, an inter-method agreement score, and a categorical certainty tier (`high` / `low` / `single`) are derived. Agreement is quantified at three levels — pairwise between methods, jointly across all methods, and per record — using the measures discussed in §4.2. A lightweight review interface lets an editor verify or correct the entries flagged as low-certainty, and the verified annotations are folded back into the TEI source with provenance distinguishing human-reviewed from automated decisions.

The interactive review component is intentionally kept light. The contribution of this paper lies in the methodological comparison and in the agreement-based triage of editorial attention (§4), not in the user interface; a fuller software description is left to a separate publication.

2.2 TEI Encoding

Annotations are written into the source register as a TEI `<sex>` child element of `<person>`, inserted after the last `<persName>` and before any `<birth>`, `<death>`, or `<note>` children. We deliberately encode the result as an element rather than as the shorthand `@sex` attribute on `<person>`: only an element gives us a place to attach the provenance attributes (`@evidence`, `@resp`, `@cert`) that the rest of this section relies on, and only an element can later be extended — with a date range, a source pointer, or alternative readings — without breaking existing consumers.

```
<person xml:id="S2472345">
  <persName type="reg">
    <surname>Aaron</surname>
    <forename>L.</forename>
  </persName>
  <sex value="male" evidence="assigned"
       resp="#gendertagger" cert="high"/>
</person>
```

The four attributes carry the editorial semantics of the annotation:

`@value`. The classified gender, restricted to `male` or `female`; entries the pipeline cannot resolve to one of these (`unknown`, `institution`, or empty) receive no `<sex>` element at all, so silence is meaningful and distinguishable from a positive assignment.

`@evidence`. Fixed to `assigned`, marking every annotation as a third-party historical assignment rather than a self-identification of the person referenced.

`@resp`. The agent responsible for the annotation: `#gendertagger` for entries left in the state produced by the automated pipeline, and a human-editor identifier (e. g. `#fko`) for entries that were verified or corrected through the review interface. The same TEI mechanism that records human responsibility statements is reused, redirected at software where appropriate.

`@cert`. A two-level confidence band, `high` or `low`. Human-reviewed entries are always `high`; automated entries inherit the certainty tier produced by the aggregation step (§4), with the unanimous/strong agreement tiers mapped to `high` and the single-method tier mapped to `low`.

This encoding is not novel in itself — `<sex>`, `@cert`, and `@resp` are standard TEI mechanisms — but applying them systematically to machine-generated annotations makes the resulting register filterable: downstream queries can target *all entries classified female with high confidence* or, equally, *all uncertain entries requiring human review*. Machine annotations cease to be invisible and become first-class, contestable editorial decisions.

3 Annotation Pipeline

The annotation pipeline combines five methodological families of automated gender classification, instantiated by nine independent classifiers in the current configuration (eight of which contribute to the combined label by default; HIVOTO-XGBoost is treated as experimental and reported in the agreement analysis but excluded from the vote):

- authority-data lookup (GND),
- dictionary-based name lookups (gender-guesser, Wiki-Gendersort, HIVOTO),
- statistical name models (nomquamgender, HIVOTO-XGBoost),
- transformer-based name classifiers (DistilBERT, BERT-genderize), and
- a locally hosted large language model (gpt-oss:20b via Ollama).

Each family encodes different assumptions about what *a name* is and what evidence licenses a gender assignment. Authority lookups are deterministic and identifier-bound but cover only a fraction of any historical register. Dictionary methods are interpretable and fast but inherit the geographic and temporal bias of their underlying name lists. Statistical models generalise to unseen forms but flatten the distribution of historical spellings into a smoothed surface. Transformer classifiers learn sub-word regularities that are robust to spelling variation but opaque to inspect. The LLM, finally, is the only method that can read the surrounding biographical context. Combining the families is therefore not just a way to boost accuracy: the agreement, or disagreement, between families becomes itself an editorial signal (§4).

3.1 Authority-Data Lookup

GND. For entries that carry a GND identifier, we resolve the gender field of the German Integrated Authority File (*Gemeinsame Normdatei*) record. The lookup is deterministic and identifier-bound: it does not classify a name, it retrieves a curated assignment. This makes it the most trustworthy single signal in the pipeline and the natural reference against which the name-based methods are evaluated. Coverage on the Schleiermacher register is approximately 30% of entries.

3.2 Dictionary-Based Name Lookups

These methods reduce a forename to a key and look it up in a precompiled name list, returning either a categorical label or a frequency-derived assignment.

gender-guesser. A widely used Python package built around a large multilingual first-name dictionary, returning one of six categorical labels (`male`, `mostly_male`, `andy`, `mostly_female`, `female`, `unknown`). Multi-token forenames are resolved to a single label by token consensus.

Wiki-Gendersort. A name-to-gender mapping derived from Wikipedia and Wikidata biographies, distributed as a flat-file lookup of several hundred thousand entries [**?**]. Included as a community-curated complement to the curated dictionaries.

HIVOTO. The *Historisches Vornamentool*, a German-language historical first-name database that records both the original spelling and a normalised classification per name, with a categorical assignment and a certainty tier. Of the dictionary methods this is the only one targeted at historical forms; on a register of late-eighteenth and nineteenth-century German correspondents it is the most directly applicable lookup we have.

3.3 Statistical Name Models

nomquamgender. A probabilistic classifier returning, per name, the estimated probability $p(\text{female})$; we threshold the probability into a categorical label and retain p as the confidence score. Van Buskirk reports accuracy comparable to commercial paid services [**?**].

HIVOTO-XGBoost. A character-n-gram XGBoost model we trained on the HIVOTO database (n-grams of length 2–5). The model gives the dictionary method's coverage a generative extension: where the lookup fails on an unseen historical spelling, the n-gram model can still produce a probability from sub-word evidence learned on the same data.

3.4 Transformer-Based Name Classifiers

Both transformer methods consume a forename string token-by-token and emit a softmax distribution over `male`/`female`.

DistilBERT (`padmajabfrl/Gender-Classification`). A DistilBERT model fine-tuned on a large name corpus for binary gender classification.

BERT-genderize (`imranali291/genderize`). A BERT model fine-tuned for the same task, included alongside the DistilBERT model so that within-family disagreement is observable rather than hidden behind a single representative.

3.5 Locally Hosted Large Language Model

gpt-oss:20b. An open-weight 20-billion-parameter LLM run locally via Ollama. Unlike all other methods, it is given the full register entry — surname, forename(s), vital dates, biographical note, alternative name forms — and asked to return a structured JSON object with a label drawn from {`male`, `female`, `organisation`, `unknown`}, a confidence value, and a free-text reason. The biographical context lets it resolve cases that defeat the name-only methods (initials with disambiguating dates, institutional names, non-German forms identified through their note text); the same context is also why it disagrees systematically with the rest of the bank, an effect we return to in §4.2.

4 Aggregation and Inter-Method Agreement

4.1 Combination Strategy

The methods are run in parallel rather than as a strict cascade, and their per-method labels are combined in three explicit steps: normalisation, voting (with two override rules), and certainty assignment. By default eight of the nine methods enter the combination — the experimental HIVOTO-XGBoost model is held out so that its agreement with the remainder can be measured cleanly. Of those eight, two (the parenthetical-qualifier rule and the GND authority lookup) are not pooled into the majority vote but applied as overrides on top of it; the actual majority count is therefore taken over the remaining six methods. The combination is unweighted: per-method confidence values are recorded alongside the labels for the editor's benefit but do not influence the vote.

Normalisation. The methods do not share a common label vocabulary — gender-guesser distinguishes `male`/`mostly_male`/`andy`, HIVOTO returns a German-language certainty tier alongside the label, the LLM emits `organisation`, and the statistical models return probabilities. Before voting, every per-method label is mapped onto a shared four-value alphabet {`male`, `female`, `institution`, `unknown`}: leaning labels (`mostly_male`, `mostly_female`) collapse to their definite counterpart, all genuinely ambiguous outputs (`andy`, `conflict`, `uncertain`) collapse to `unknown`, and absent or unparseable answers are recorded as abstentions rather than as `unknown` votes. The distinction matters: an abstention means the method had no input to work with (e. g. an entry without a forename), whereas `unknown` means the method considered the input and could not decide.

Voting. The combined label is the plain (unweighted) majority of the substantive votes across the six pooled methods that did not abstain, with one tie-breaking rule: when `unknown` ties with `male`, `female`, or `institution`, the substantive label wins, since `unknown` encodes "no information" rather than positive evidence against the substantive option. Two authority overrides take precedence over the vote when applicable: a parenthetical-qualifier rule that scans the surname, forename, and biographical note for German kinship markers (e. g. *Frau des* . . . , *Sohn der* . . .) and structural disambiguators in parentheses; and a non-`unknown` GND lookup result. When either override fires, its label replaces the majority verdict and the entry's certainty

is forced to `high` on the grounds that both signals are deterministic (rule-based or authoritative) rather than probabilistic. The agreement score recorded for the entry is the share of votes carried by the winning label among the methods that did vote.

Certainty assignment. From the agreement score and the number of voting methods, each entry is assigned a categorical certainty tier: `high` when at least two methods voted and at least three quarters of them agreed (or whenever an authority override fired); `low` when at least two voted and a majority of half or more agreed; `contested` when at least two voted but agreement fell below half; `single` when only one method had something to say and that lone method was the LLM with self-reported `high` confidence; `uncertain` when the lone method was the LLM with `low` or `uncertain` self-reported confidence; and `no_data` when no method could vote. These tiers — not the raw agreement fraction — are what the TEI `@cert` attribute (§2.2) ultimately records.

On the Schleiermacher register, the resulting tier distribution is heavily concentrated at `high` (Table 1): 6,934 entries (83.8%) reach the high-confidence tier through majority agreement or an authority override, while only 295 entries (3.6%) fall into the genuinely problematic `low` or `contested` tiers. The remaining 1,048 entries (12.7%) are LLM-only cases (`single` / `uncertain`) — entries on which only the LLM, with its full biographical context, was able to commit to a label.

`@cert` tier	Entries	Share
`high`	6,934	83.8%
`single`	710	8.6%
`uncertain`	338	4.1%
`low`	257	3.1%
`contested`	38	0.5%

Table 1: Distribution of the combined certainty tier across the Schleiermacher register (N = 8,277).

We deliberately did not collapse the methods into a single black-box ensemble. The per-method labels are kept alongside the combined result in the aggregated table, both because editors gain useful epistemic signal from the agreement pattern — nine methods agreeing on *female* is qualitatively different from five against three with the LLM dissenting — and because the labels can be re-aggregated under a different combination strategy without re-running the underlying classifiers. The latter matters in practice for the slow methods: re-running the local LLM over several thousand entries takes hours, while changing the voting rule takes seconds.

4.2 Inter-Method Agreement

To quantify how strongly the methods agree we report three complementary measures, computed on the normalised label projection introduced above and reduced for the agreement analysis to the three categories {`male`, `female`, `ambiguous`}, with abstentions treated as missing.

Cohen's κ is reported pairwise between every pair of methods, restricted to the entries on which

both methods have a substantive label (and excluding pairs sharing fewer than ten entries). Cohen's κ is the standard two-rater chance-corrected agreement coefficient and gives an interpretable matrix view of which methods cluster together and which sit apart.

Fleiss' κ generalises the chance-corrected view to an arbitrary fixed number of raters with categorical labels and is reported as a single scalar over the subset of entries on which *all* methods returned a substantive label. It answers the question how consistent the bank as a whole is on the entries where it has full coverage.

Krippendorff's α complements Fleiss' κ by tolerating missing values without restricting the analysis to fully covered entries, and is reported as a second scalar over the full register. The two measures together separate the agreement *when everyone speaks* (Fleiss') from the agreement *across the whole register including abstentions* (Krippendorff).

The pattern that emerges on the Schleiermacher Digital register is structured rather than uniform. Across the methods, the pairwise Cohen's κ values cover the range $[0.56, 0.89]$ with an unweighted average of 0.75; Fleiss' κ over the 5,984 entries on which all methods returned a substantive label is 0.77, and Krippendorff's α over the full register of 8,277 entries is 0.75. The pairwise matrix (Figure 1, with numerical values in Table 2) shows three internally tight clusters. The two HIVOTO methods agree most strongly with one another ($\kappa = 0.89$) and both agree closely with nomquamgender ($\kappa = 0.88$, 0.88); gender-guesser, Wiki-Gendersort, and nomquamgender form a second tight cluster of dictionary-and-statistical methods (κ between 0.82 and 0.87); and the two transformer classifiers agree at $\kappa = 0.80$. The LLM sits apart from all three clusters, agreeing with the name-only methods at noticeably lower values: κ between 0.62 and 0.77 across the eight pairs, with the lowest agreement against the BERT-genderize transformer ($\kappa = 0.56$). This is a property of the method rather than a defect: the LLM is the only classifier reasoning from biographical context rather than from name strings alone.

	HV	HV-XGB	NQG	GG	WGS	HF-gz	LLM	HF-DB
HIVOTO	—	0.89	0.88	0.85	0.86	0.63	0.73	0.69
HIVOTO-XGB		—	0.88	0.82	0.83	0.67	0.77	0.73
nomquamgender			—	0.87	0.82	0.62	0.75	0.69
gender-guesser				—	0.87	0.64	0.70	0.71
Wiki-Gendersort					—	0.70	0.70	0.76
HF BERT-gz						—	0.56	0.80
LLM							—	0.62
HF DistilBERT								—

Table 2: Pairwise Cohen's κ values underlying Figure 1. HV = HIVOTO, NQG = nomquamgender, GG = gender-guesser, WGS = Wiki-Gendersort, HF-gz = HuggingFace BERT-genderize, HF-DB = HuggingFace DistilBERT.

Per-record agreement. The aggregate κ values quantify how much the methods agree *on average*, but for editorial triage we need to know on *which* entries they disagree. For every register entry we therefore additionally compute Fleiss' P_i, the per-item agreement component

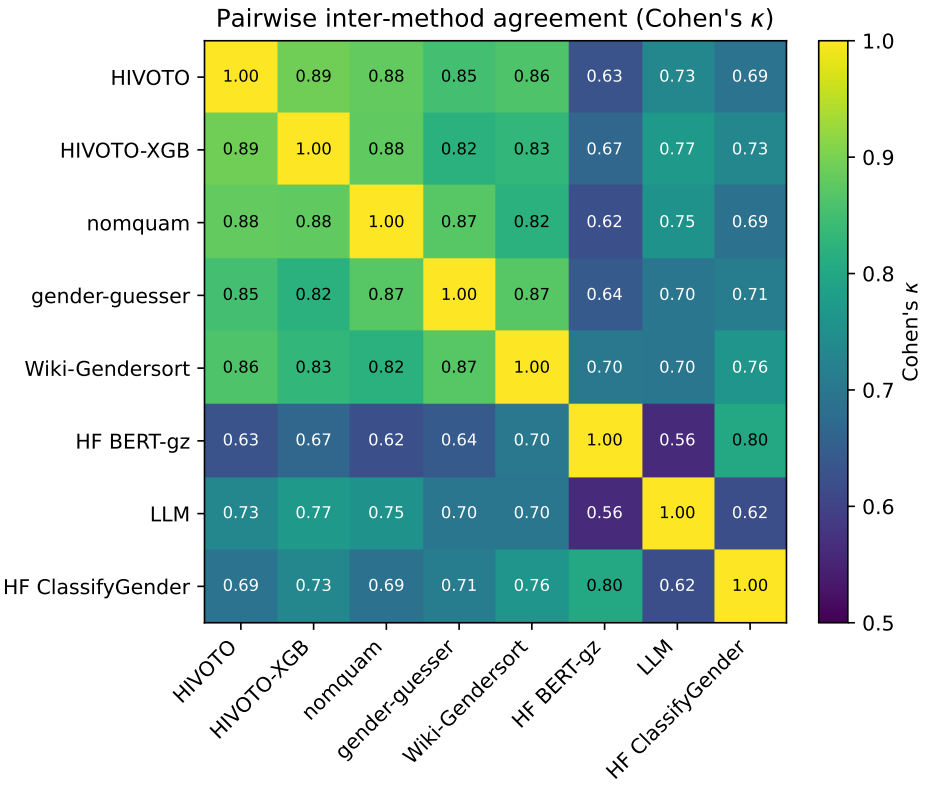


Figure 1: Pairwise inter-method agreement (Cohen's κ) on the Schleiermacher Digital register. Three internally tight clusters are visible (HIVOTO/HIVOTO-XGB/nomquamgender, the dictionary methods gender-guesser/Wiki-Gendersort, and the two transformers); the LLM sits apart from all three. GND and the parenthetical-qualifier rule are excluded because their coverage overlap with the name-based methods is too small for a stable pairwise estimate.

of Fleiss' κ, together with the Shannon entropy H_i of the label distribution across the methods. Both summarise the same underlying vote distribution: P_i measures how concentrated the votes are on a single label (1.0 = unanimous, lower bound = uniform across labels), and H_i measures how dispersed they are. From these two values each entry is sorted into one of six tiers — `unanimous`, `strong_majority` (off by at most one method), `weak_majority`, `no_consensus`, `single_rater`, and `no_answer` — which are exactly the tiers exposed to the editor in the review interface as a triage signal.

On the Schleiermacher register, this triage is heavily right-skewed: of the 8,277 entries, 4,441 (53.7%) fall into the `unanimous` tier and a further 2,982 (36.0%) into `strong_majority`. Only 492 entries (5.9%) land in `weak_majority` and 362 (4.4%) in `no_consensus` — together 854 entries (roughly one in ten) on which the bank cannot agree, and which the review interface surfaces first. The skew is also informative by majority label (Table 3): where the methods settle on `male`, 80.8% of entries are unanimous (mean $P_i = 0.93$); where they settle on `female`, 56.8% are unanimous (mean $P_i = 0.84$); and entries the bank can only push

into `ambiguous` are never unanimous (mean $P_i = 0.40$). The asymmetry between `male` and `female` agreement is itself a finding we return to in §[TODO: discussion section]. Figure 2 visualises the same data as a stacked-bar breakdown.

Majority label	N	unanimous	strong_maj.	weak_maj.	no_consensus
`male`	4,548	3,677 (80.8%)	524 (11.5%)	216 (4.7%)	131 (2.9%)
`female`	1,346	764 (56.8%)	314 (23.3%)	213 (15.8%)	55 (4.1%)
`ambiguous`	240	0 (0.0%)	5 (2.1%)	59 (24.6%)	176 (73.3%)

Table 3: Per-record agreement tier broken down by the majority label. The methods agree much more readily on `male` than on `female`; entries pushed to `ambiguous` are never unanimous and dominate the editorial review queue. (The remaining 2,143 entries with $n_{\text{answered}} \leq 1$ fall into `single_rater` or `no_answer` and are omitted.)

For the entries that fall into `no_consensus` and `weak_majority`, manual inspection of the most-contested records shows that disagreement clusters around four recurring profiles:

1. Institutional names mis-parsed as personal names (e. g. corporate correspondents).
2. Initials-only entries, where dictionary methods have nothing to match against.
3. Non-German names lying outside the coverage of the dictionary-based classifiers.
4. Historically ambiguous names whose gendered usage shifted between the eighteenth and twenty-first centuries — a problem that modern training data inherently amplifies.

These are precisely the entries flagged for editorial review: disagreement functions as a triage signal, directing editor attention to cases where automated annotation is least reliable.

4.3 Output Categories

genderTagger uses four output categories: `male`, `female`, `uncertain`, and `institution`. The binary main classification reflects what name-based methods can in principle resolve — a statistical association between historical name forms and recorded gender. Non-binary, trans, and agender identities are not addressable from name strings alone and are deliberately not represented as classifier output categories. This is a limit we record rather than work around: the tool documents what it does not know, in the same spirit that `@cert` records what it is uncertain about. We return to this question in §[TODO: discussion / limitations section].

5 Case Study: Schleiermacher Digital

The development of genderTagger was conducted in collaboration with the *Schleiermacher Digital* edition, which provides digital access to the correspondence, lectures, and diaries of the theologian Friedrich Schleiermacher (1768–1834).

The register. The edition's person register contains 8,277 entries at the time of writing. Only 2,253 of those entries (27.2%) carry a GND identifier resolvable to a gendered authority record; the remaining 6,024 entries (72.8%) require alternative classification approaches. This distribution is representative of the practical limits of authority-file-based enrichment for a register of this size and historical scope: even a relatively well-curated GND attachment leaves nearly three quarters of the entries without an authoritative gender signal, motivating the multi-method approach described in §4.

Coverage per method. Coverage on the Schleiermacher register varies systematically with each method's input requirements (Table 5). The local LLM, which receives the full register entry including biographical note, is the only method that produces a label for every entry. The five name-based classifiers — gender-guesser, the two transformers, HIVOTO, and nomquamgender — each cover the same 6,134 entries (74.1%): the gap to full coverage is exactly the set of entries with no usable forename. Wiki-Gendersort covers a slightly smaller subset (73.4%) because its dictionary is more restrictive on non-Latin script and rare forms. The parenthetical-qualifier rule reaches 33.5% of entries — almost entirely because Schleiermacher's correspondents include a substantial number of women identified in the source through their relationship to a male principal (*Frau des . . .* , *Schwester von . . .*), a pattern the rule captures explicitly.

Per-method label distributions. Coverage tells us only how often each method commits to *some* label; the per-method label distribution (Table 4) shows *which* labels each method tends to assign, and exposes systematic bias profiles that the aggregate κ values smooth over. Three patterns are worth highlighting. First, the GND-matched subset is itself strongly skewed: of the 2,253 entries with a usable GND record, 90.9% resolve to `male` and only 9.1% to `female` — a more extreme imbalance than any name-based method produces. Second, the two HuggingFace transformers diverge markedly on the same 6,134-entry coverage set: HF BERT-genderize predicts $\sim$5$\times$ as many `male` as `female` entries (56.2% vs. 11.2%), while HIVOTO on the identical input predicts only $\sim$3$\times$ as many (53.3% vs. 17.6%). The HF Genderize bias toward `male` is the dominant source of its low κ against the LLM (Table 2). Third, the parenthetical-qualifier rule is the only method whose label distribution is approximately gender-balanced (13.7% `male` vs. 14.0% `female`); this reflects the fact that it fires almost exclusively on relational descriptors, which in this corpus are predominantly used to identify women.

Per-method accuracy against GND. Treating the 2,253 entries with a gendered GND authority record as a benchmark set allows a direct comparison of per-method classification quality (Figure 4). Effective accuracy — correct labels divided by the full benchmark size, so abstentions count against the method — ranges from 86.5% (the two transformer classifiers, which abstain on $\sim$12% of the benchmark) to 99.7% (the local LLM, which answers on every entry). The more revealing axis is the imbalance between male and female recall. The two transformer classifiers achieve 88–89% on male names but only 60–71% on female names, the largest per-class gaps in the cohort; gender-guesser, Wiki-Gendersort, HIVOTO, and nomquamgender close the gap considerably (85–91% female recall against 88–90% male); only the LLM is balanced across classes (99.7% male, 100% female on the GND benchmark). The benchmark itself is

Method	male	female	inst.	unknown	abstain
GND	2,048 (24.7%)	205 (2.5%)	0 (0.0%)	0 (0.0%)	6,024 (72.8%)
Parenth. qualifier	1,133 (13.7%)	1,159 (14.0%)	7 (0.1%)	472 (5.7%)	5,506 (66.5%)
gender-guesser	4,332 (52.3%)	1,278 (15.4%)	0 (0.0%)	524 (6.3%)	2,143 (25.9%)
Wiki-Gendersort	4,457 (53.8%)	1,292 (15.6%)	0 (0.0%)	326 (3.9%)	2,202 (26.6%)
HIVOTO	4,413 (53.3%)	1,457 (17.6%)	0 (0.0%)	264 (3.2%)	2,143 (25.9%)
nomquamgender	4,379 (52.9%)	1,380 (16.7%)	0 (0.0%)	375 (4.5%)	2,143 (25.9%)
HF BERT-gz	4,653 (56.2%)	926 (11.2%)	0 (0.0%)	555 (6.7%)	2,143 (25.9%)
HF DistilBERT	4,569 (55.2%)	1,117 (13.5%)	0 (0.0%)	448 (5.4%)	2,143 (25.9%)
LLM	5,426 (65.6%)	1,897 (22.9%)	119 (1.4%)	835 (10.1%)	0 (0.0%)
Combined	5,452 (65.9%)	1,867 (22.6%)	83 (1.0%)	875 (10.6%)	0 (0.0%)

Table 4: Per-method label distributions on the Schleiermacher register ($N = 8{,}277$). Percentages are over the full register (so `abstain` = 1 minus the coverage in Table 5). The combined row is the output of the aggregation pipeline (§4).

Method	Coverage	Share
Local LLM (gpt-oss:20b)	8,277	100.0%
gender-guesser	6,134	74.1%
HIVOTO	6,134	74.1%
HF DistilBERT (padmajabfrl)	6,134	74.1%
HF BERT-genderize (imranali291)	6,134	74.1%
nomquamgender	6,134	74.1%
Wiki-Gendersort	6,075	73.4%
Parenthetical-qualifier rule	2,771	33.5%
GND authority lookup	2,253	27.2%

Table 5: Per-method coverage on the Schleiermacher Digital person register ($N = 8{,}277$). Coverage is the number of entries on which the method returned a substantive label (not an abstention).

small and male-dominated (2,048 male vs. 205 female), so female-recall figures should be read as ordinal rather than absolute, but the ordering is consistent with the disagreement profiles in §4.2: methods with weaker female recall are also the methods that contribute disproportionately to the contested entries on female names.

Where methods disagree. The disagreement profiles introduced in §4.2 appear at meaningful scale on this register. Of the 8,277 entries, 854 (10.3%) fall into the `weak_majority` or `no_consensus` tiers and constitute the editorial review queue. Manual inspection of the most-contested entries shows that the four disagreement profiles — institutional names mis-parsed as personal names, initials-only entries, non-German forms, and historically ambiguous names — are not evenly distributed: the top contested records are dominated by entries where the source itself encodes a relationship rather than an independent identity (e. g. *(Frau des F.)*, *(Sohn der Manon)*) and by initials-only references that defeat every name-based method.

Resulting gender distribution. The combined classification yields a strongly imbalanced distribution: of the 8,277 entries, 5,452 (65.9%) are labelled `male`, 1,867 (22.6%) `female`,

83 (1.0%) `institution`, and 875 (10.6%) remain `unknown`. The male-to-female ratio of roughly 2.9:1 is itself a finding about the source rather than about the method: a private and academic correspondence of an early-nineteenth-century Berlin theologian skews structurally toward male correspondents (university colleagues, ecclesiastical and political officeholders, publishers). The agreement evidence corroborates this: the methods agree unanimously on 80.8% of the `male` entries but only on 56.8% of the `female` entries (§4.2), suggesting that the `female` category is both smaller in absolute terms and harder to assign reliably — in part because women in this corpus are disproportionately referenced through relational descriptors that name-based methods cannot resolve.

Lessons for comparable editions. The case study suggests three transferable observations for editorial projects in a similar position. First, GND coverage alone is unlikely to suffice for any nineteenth-century or earlier register of comparable size; the long tail of non-authority entries demands a fallback strategy. Second, multi-method agreement provides not only a label but a meaningful triage signal: the $\sim$10% of entries on which the methods cannot agree concentrates exactly the cases where editorial attention is most warranted. Third, source-specific structural patterns — in this register, the relational naming convention for women — can be addressed by lightweight, edition-specific rules (the parenthetical-qualifier method here), provided the combination strategy is open enough to admit them as full peers of the off-the-shelf classifiers.

6 Human-in-the-Loop Workflow

Rather than treating automated classifications as definitive assignments, genderTagger is explicitly designed as a human-in-the-loop workflow in which editorial decisions remain subject to human interpretation and revision. A typical session runs: import TEI $\rightarrow$ batch-classify with all methods $\rightarrow$ inspect agreement and disagreement $\rightarrow$ annotate (confirm or override) $\rightarrow$ export. Classifier execution is decoupled from the user interface: long-running methods such as transformer models and the local LLM run in the background, while editors can already begin reviewing entries that the faster methods have classified. This matters in practice — running a 20-billion-parameter local LLM over several thousand register entries takes hours, and a blocking UI would make the tool unusable on registers of realistic size.

7 Conclusion and Future Work

genderTagger demonstrates how semi-automated gender annotation can support metadata enrichment workflows in TEI/XML-based digital editions while preserving editorial oversight and interpretative responsibility. By combining multiple classification methods, integrating uncertainty-aware review mechanisms, and embedding annotation directly into existing editorial infrastructures, the system contributes both methodological and infrastructural perspectives to ongoing discussions surrounding bias, metadata enrichment, and human-centered AI in the Digital Humanities.

Future work will focus on expanding multilingual support, improving explainability and confidence estimation, integrating additional authority data sources such as Wikidata, and exploring

adaptive learning approaches based on editorial feedback.

Acknowledgements: KD acknowledges support by DFG Project 000000000.

Data availability: My data is publicly available on zenodo.

Declaration on the use of AI: No AI hallucinations over here.

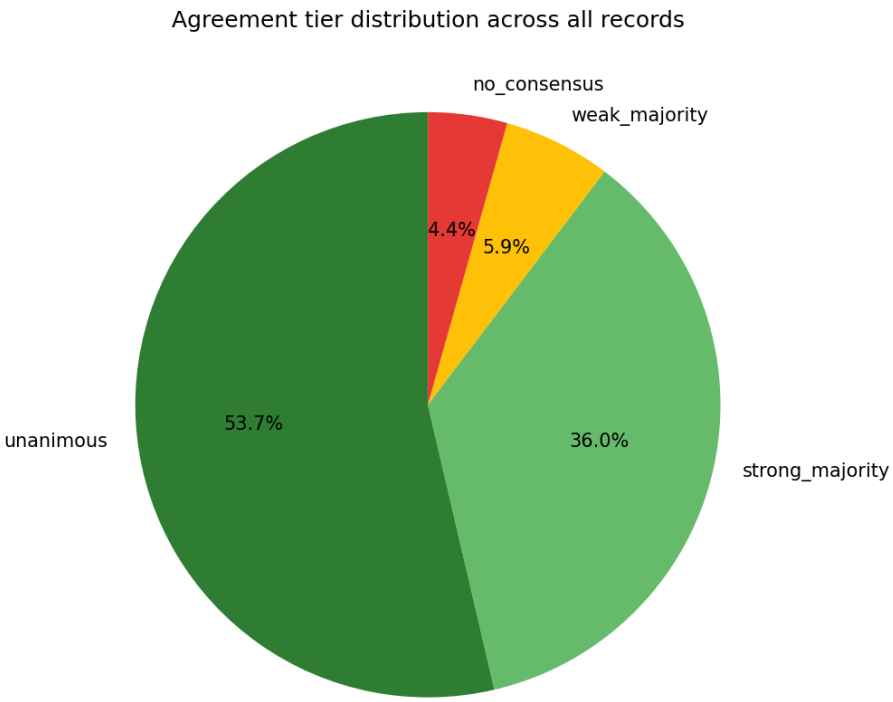


Figure 2: Per-record agreement tiers, segmented by the majority label assigned by the bank. The yellow `unanimous` band dominates the `male` column; the `female` column is more evenly split across tiers; and `ambiguous` entries cluster in the lower (less-agreed) tiers. The P_i and H_i distributions underlying these tiers are shown in Figure 3.

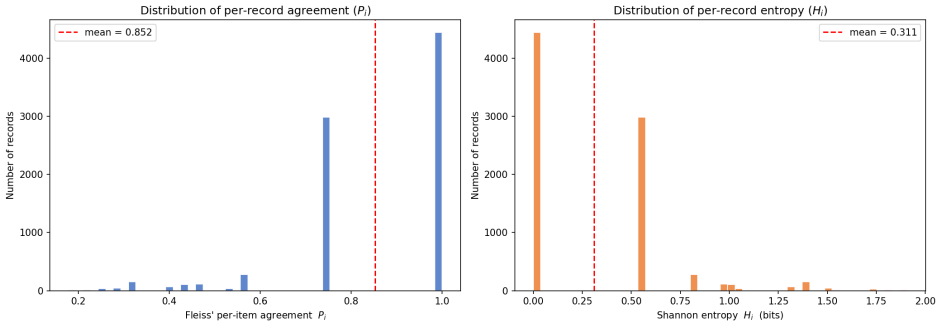


Figure 3: Distributions of Fleiss' per-item agreement P_i (left) and Shannon entropy H_i (right) across the 8,277 register entries. Both distributions are strongly bimodal: most entries are close to unanimity ($P_i \to 1$, $H_i \to 0$), with a long thin tail of contested entries that drives the editorial review queue.

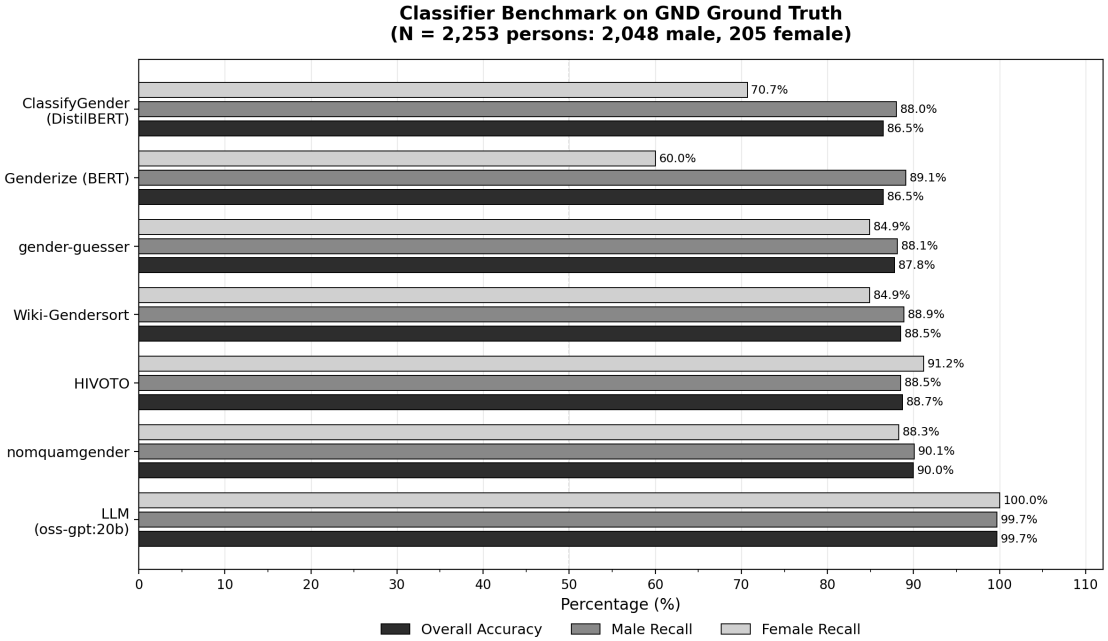


Figure 4: Per-method effective accuracy and per-class recall on the Schleiermacher GND benchmark subset ($N = 2{,}253$ persons; 2,048 male, 205 female). Effective accuracy counts abstentions as errors, which is why the transformer classifiers — despite high precision when they do answer — score below the dictionary methods. The female-recall column shows the largest spread between methods.

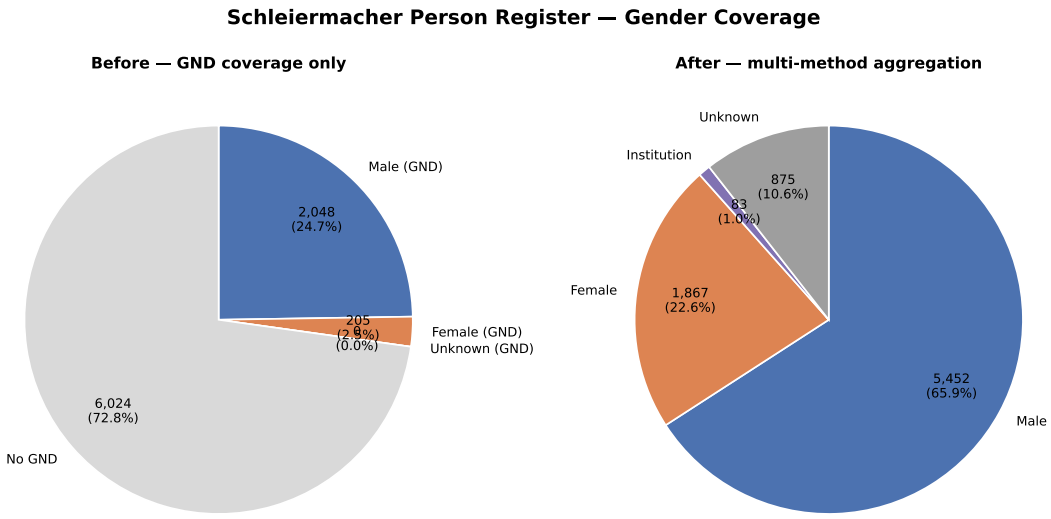


Figure 5: Gender coverage of the Schleiermacher person register ($N = 8{,}277$) before and after multi-method aggregation. *Left*: GND-only baseline — 72.8% of entries have no resolvable authority record. *Right*: combined output of the aggregation pipeline (§4); every entry receives a label, with the residual `unknown` category restricted to entries on which the methods either abstained or could not reach consensus.